5

Reasoning and Learning in Artificial Intelligence

5.1 Reasoning

The primary aim of any artificially intelligent system is to think like humans and mimic their actions to perform tasks faster in a smarter fashion. In this process, reasoning plays a vital role as it helps draw conclusions, make predictions and construct approaches towards a particular thought with the help of existing knowledge.

The term 'reasoning' is associated with thinking, intellect and cognition that directs the process of generating one idea from the other. It involves analysing the given information, comparing it with what is known and then coming to a conclusion. Therefore, reasoning is the key to making decisions, solving problems and evaluating actions. For example, Siri (virtual assistant) uses reasoning to offer suggestions and recommendations based on commands like nearest location, the date for tomorrow, etc.

5.1.1 Features of a Reasoning System

The four qualities of a reasoning system are discussed as follows:

1. A reasoning system is **a self-contained model or system** as it is not modified externally by a programmer or user for carrying out the tasks at hand. It is a collection of models or algorithms or systems that is untouched in the course of learning new tasks.
2. A reasoning system can solve a different set of problems either through examples, or instruction or even trial-and-error. Some of these problems may be new, unique and unusual.
3. It gets better through a limited number of trials through practice.

4. A reasoning system should solve new problems in a limited time frame and from a small number of examples.

5.1.2 Why Focus on Reasoning System?

Before reading about why to go for a reasoning system, let us first try to understand the difference between prediction, inference and reasoning.

Prediction is done by pattern matching. A prediction system accepts a lot of data and then maps input to output to predict outputs.

Inference is the process of generating new information. It is an automatic process that has no goals. An inference system can be thought of as a machine that takes some inputs and processes it to derive new information as output.

Reasoning system, on the other hand, is targeted or is directed to achieve the goal of deducing an inference. For this, inputs are carefully selected, organized and processed to generate the desired result.

Machines use reasoning, along with knowledge representation, logic and learning for analysis, problem-solving and making conclusions.

Nowadays, extensive research has been done to make machine learning algorithms meticulously perform prediction tasks when the problem is well-defined and a large amount of data is available. However, these systems are prone to adversarial attacks. To improve their capabilities, AI systems must themselves identify and recover from failure, and solve problems without any external help. Let us now try to understand the different methods of reasoning.

5.1.3 Methods of Reasoning

We know that reasoning is the mental process of deriving a logical conclusion and making predictions from available knowledge, facts and beliefs. It helps us to derive valid conclusions by thinking rationally to infer facts from existing data. In artificial intelligence, reasoning facilitates ma-

chines to think rationally and perform like a human. Reasoning is classified into the following types:

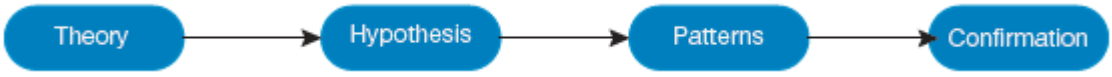
Deductive Reasoning

It is a strategic approach that uses available facts, information or knowledge to draw valid conclusions. Every result is based on facts and ideas and is drawn using a top-down approach. In deductive reasoning, the arguments can be valid or invalid based on the value of the premises (rules or facts). A conclusion is true only if the value of the premises is true.

Deductive reasoning scans a generalized statement to provide a valid conclusion. For example:

- Young people in the age group 15 –25 are active on social media.
- Out of 60 students in a class, the ratio of boys is more than that of girls.

Thus, deductive reasoning uses formal logic to produce logically certain results. This process of drawing conclusion involves different steps described as follows.



Deductive reasoning usually starts from the general premises to get to specific conclusion(s). Compared to inductive reasoning (discussed below), it is difficult to use deductive reasoning as it requires true facts. For example:

Premise-1: All humans breathe.

Premise-2: Naina is a human.

Conclusion: Naina breathes.

Inductive Reasoning

Contrary to deductive reasoning, inductive reasoning also known as cause-effect reasoning; it generates hypothesis rather than drawing con-

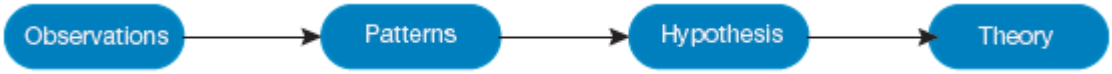
clusion(s) to the facts at the beginning of the process.

Inductive reasoning follows a bottom-up approach to make generalized statements from specific facts and knowledge. Therefore, there may be a case when the premises are true but the conclusion may not be true as the inductive argument may be either strong or weak. Some examples of inductive reasoning are as follows:

- All the students scoring 100% are from state X.
- City X recorded the lowest temperature in north India.

Reasoning allows AI technologies to extract critical information from large sets of structured and unstructured data.

Inductive reasoning is preferred in cases with limited set of facts and data to deduce a conclusion. In such a scenario, the inductive reasoning technique uses historical data to produce a generic rule whose conclusion is supported by the premises. Therefore, it is said to be exploratory in nature as it allows for uncertain but likely results. The steps involved in inductive reasoning include the following:



Therefore, inductive reasoning is fast and easy, as it requires evidence instead of facts. It can derive conclusions using limited data by the process of generalization (start with specific data to a general statement or conclusion). A comparative analysis of inductive and deductive reasoning is given in **Table 5.1**.

TABLE 5.1 Inductive vs Deductive reasoning

Attribute	Deductive Reasoning	Inductive Reasoning
Definition	Deductive reasoning is a form of valid reasoning that deduces new information from known related facts and information.	Inductive reasoning arrives at a conclusion by the process of generalization using specific facts or data.
Approach	It follows a top-down approach.	It follows a bottom-up approach.
Starts from	It starts from the premises (or rules).	It starts from the conclusion.
Validity	Conclusion must be true if the premises are true.	Truth of premises does not guarantee the truth of conclusions.
Usage	It is difficult to use as facts that must be true are required.	It is fast and easy to use, as we need evidence instead of true facts. It is mostly used in our daily lives.
Process	Theory → hypothesis → patterns → confirmation.	Observations → patterns → hypothesis → theory.
Argument	Arguments may be valid or invalid.	Arguments may be weak or strong.
Structure	It reaches from general facts to specific facts.	It reaches from specific facts to general facts.
Conclusion	The conclusions are certain.	The conclusions are probabilistic.

Common Sense Reasoning

It is the most frequently used reasoning in daily life and usually comes from day-to-day experiences. When we humans face a different situation in life, we gain some knowledge. If, in the future, we encounter the same situation, we use our past experiences to draw a conclusion to handle that situation. Some of the examples are as follows:

- If, in a lockdown, we went out of our houses without a mask and had to pay a heavy fine, then the next time, we go out of the house, we are aware about consequences and will not repeat the same action.
- While crossing roads, we always recall a set of rules to be followed.

Inductive and deductive reasoning are the forms of propositional logic.

Common sense reasoning helps machines simulate human ability to make decisions and perceptions about real-world events. In AI systems, it is mainly used in applications involving computer vision and robotic manipulation.

1. Common sense reasoning is an informal form of reasoning, which can be gained through experiences.
2. It simulates the human ability to make presumptions about events which occurs on every day.
3. It relies on good judgment rather than exact logic
4. Common sense works with **heuristic knowledge** and **heuristic rules**.

Any theorem proving is an example of monotonic reasoning.

Monotonic Reasoning

Monotonic reasoning uses facts, information and knowledge to draw a conclusion. However, once a conclusion is taken, it always remains constant. Addition of new facts and knowledge in the database cannot alter the conclusion drawn at an earlier point in time. It is mainly used in conventional reasoning systems and logic-based systems. Examples of monotonic reasoning are as follows:

- The Harappan civilization was located in the Indus river valley.
- Kashmir is hailed all over the world for its incredible natural beauty.

Human beings' perceptions for various things in daily life is an example of non-monotonic reasoning.

However, since monotonic reasoning does not modify the conclusion based on the new data, it cannot be used in systems that accept real-time inputs to generate outputs.

Monotonic reasoning is used in conventional reasoning systems, and a logic-based system is monotonic.

Advantages

1. Each proof will always remain valid.
2. Facts deduced from existing facts will always remain valid.

Abductive reasoning is a form of logical reasoning which starts with observation(s) to find the most likely explanation or conclusion for the observation.

Disadvantages

1. Real-world scenarios cannot be expressed using monotonic reasoning.
2. Hypothesis knowledge cannot be expressed. Monotonic reasoning represents only true facts.
3. It allows deriving conclusions from the old proofs, so new knowledge from the real world cannot be added.

Non-Monotonic Reasoning

Unlike monotonic reasoning, non-monotonic reasoning labels a conclusion as invalid if on addition of new or more information, a previously generated conclusion becomes contradictory. This type of reasoning usually deals with uncertain and incomplete models.

Non-monotonic logics are devised to capture and represent defeasible inferences (draw tentative conclusions). These conclusions can be retracted when further evidences are available.

Consider the following example. If we have, in our knowledge base, statements like, ‘Typically, birds fly. Penguins do not fly. Tweety is a bird’. It is plausible to conclude that Tweety flies. However, if at a later point in time, we add in the knowledge base that Tweety is a penguin, then the previous conclusion must be retracted and a new conclusion will be drawn that would say, Tweety does not fly.

Advantages of Non-Monotonic Reasoning

1. Non-monotonic reasoning is very useful for real-world systems such as robot navigation.
2. It helps to choose probabilistic facts or can make assumptions.

Disadvantages of Non-Monotonic Reasoning

1. Old facts may be invalidated by adding new sentences.
2. It cannot be used for theorem proving.

Abductive Reasoning

This type of reasoning is quite different from those discussed above. Abductive reasoning is an extension of deductive reasoning, and starts with an incomplete set of facts, information and knowledge to find the most appropriate explanation and conclusion. Conclusions are based on facts known at the present. It does not give any importance to outdated facts and information. Such type of reasoning is very important in making decisions in our daily lives. For example:

- A doctor draws conclusions about a patient's health based on the present test reports.
- You will not plan a vacation to a country that is currently facing a war-like situation irrespective of how beautiful and adventurous it has been in the past.

Thus, abductive reasoning, is a form of logical inference that uses theories to define observations having the simplest and likely explanation to give the most accurate suggestions and guesses. It is usually associated with troubleshooting and decision-making. However, the main drawback is that it does not rectify the plausible conclusion. So, it must be used only in case of data certainties. Moreover, abductive reasoning can justify any belief (even absurd ones). Therefore, it is not always reliable.

5.2 Probabilistic Reasoning in Artificial Intelligence

5.2.1 Uncertainty

Recall that, in first-order logic and propositional logic, knowledge is represented by certainty. This means that we were sure about the predicates. So, if we say, A→B, then it indicates that if A is true, then B is also true. But what if there is a situation where we are not sure about whether A is true or not? How will we express this statement? This situation is known as uncertainty.

To represent uncertain knowledge, where we are not sure about the predicates, we need uncertain reasoning or probabilistic reasoning. However,

before going to probability reasoning, let us first look at some causes of uncertainty in the real world.

1. Information from unreliable sources
2. Experimental errors
3. Equipment fault
4. Temperature variation
5. Climate change

In AI, we use probability to indicate uncertainty in knowledge. In probabilistic reasoning, probability theory is combined with logic to handle uncertainty. For example, in real world, the statement, 'It will rain today', is a probable sentence as we can only assume that it will happen but are not sure about it. Thus, probabilistic reasoning can be applied in the following circumstances:

- When there are unpredictable outcomes.
- When specifications or possibilities of predicates become too large to handle.
- When an unknown error occurs during an experiment.

Now, before jumping into probabilistic reasoning to solve problems with uncertain knowledge, let us recapitulate some often-used key terms.

5.2.2 Probability

Probability can be defined as a chance that an uncertain event will occur. It indicates the likelihood that an event will occur. The value of probability is between 0 and 1. Here, probability = 0 means event will never happen and probability = 1 means that event will definitely take place. Technically, P(A), that is, the probability of an event A is: $0 \leq P(A) \leq 1$

P(A) = 0, indicates total uncertainty and

P(A) =1, indicates total certainty in an event A.

The probability of an uncertain event can be expressed as

$$\text{Probability of occurrence} = \frac{\text{Number of desired outcomes}}{\text{Total number of outcomes}}$$

- P(¬A) = probability of A not happening event.
- P(¬A) + P(A) = 1.

Event: Each possible outcome of a variable is called an event. For example, when a coin is tossed, you can either get a head (event 1) or a tail (event 2).

Sample space: The collection of all possible events is called sample space.

Prior probability: The prior probability of an event is probability computed before observing new information.

Posterior probability: This refers to the probability that is calculated after all evidence or information has been collected. It combines prior probability and new information.

Independent Event: It is an event which is not affected by other events. For example, tossing a coin as each toss of a coin is not affected by any toss in the past. The probability of getting a head will always be 1-in-2, or 50%.

Dependent Events: These events can be affected by previous events. For example, if there are 2 blue and 3 red marbles in a bag, then the chances of getting a blue marble is 2 in 5. But after taking one red marble earlier, the chance of taking a blue marble next is 2 in 4. Similarly, had we taken a blue marble before, the chance of a blue marble next is 1 in 4. Since the next event depends on what happened in the previous event, it is called a dependent event.

Conditional probability is the possibility of an event based on the existence of a previous event. It is calculated by multiplying the probability of the preceding event with that of the succeeding event.

The conditional probability of event A is the probability that event A will occur given that another event B has already occurred. There are two

types of events—dependent events or independent events. Conditional probability for these events can be calculated as follows:

$$P(A \mid B) = P(A \text{ and } B) / P(B),$$

if A and B are two dependent events.

If A and B are independent events, then conditional probability is given by, $P(A \mid B) = P(A)$

Marginal probability: The unconditional probability of an event P(A) is known as marginal probability. This is not dependent on the occurrence of any other event. For example, the probability that a card drawn from a deck of cards is red (P(A) = 0.5), and the probability that the card is drawn is 8 (P(B)=1/13). Both these events are independent events since the outcome of one event is not affected by the occurrence of the other.

Joint Probability: A joint probability is the probability of event A and event B happening, P(A and B) together. Actually, it is the likelihood of the intersection of two or more events and thus, written as intersection of A and B, that is, $P(A \cap B)$. For example, the likelihood that a card drawn from a deck of cards is red and numbered 8 is equal to P(A and B) = 2/52 = 1/26. (Since we have two Black-8 in a deck of 52).

Conditional probability: Conditional probability is the probability of occurring an event when another event has already happened. This can be expressed as

$$P(A \mid B) = \frac{P(A \wedge B)}{P(B)}$$

In the above formula, we are calculating the event A when event B has already occurred. Here,

$P(\boldsymbol{A} \wedge \boldsymbol{B})$ is the joint probability of A and B

P(B)= Marginal probability of B.

Example: In a class, 70% of the students want to study R Language and 40% of the students want to learn Python Programming. What is the percent of students who want to study R as well as Python?

Solution:

Let A be an event that a student wants to study Python.

B is an event that a student wants to study R.

$$P(\text{A} \mid \text{B}) = \frac{P(\text{A} \wedge B)}{P(\text{B})} = \frac{0.4}{0.7} = 57\%$$

Hence, 57% are the students who want to study R also like Python.

5.3 Bayes' Theorem

Bayes' theorem, also known as **Bayes' rule, Bayes' law** or **Bayesian reasoning,** determines the probability of an event with uncertain knowledge.

In probability theory, Bayes' rule is related to the conditional probability and marginal probabilities of two random events. It facilitates computing the value of P(B | A) when the value of P(A | B) is known. Moreover, the probability prediction of an event can be easily updated when new information is added.

For example, if cancer corresponds to one's age, then we can determine the probability of cancer in a patient more accurately by using Bayes' theorem.

Bayes' theorem can be derived using product rule and conditional probability of event A with known event B. Thus, we can write:

$$P(A \wedge B) = P(A \mid B)\ P(B) \qquad \text{(i)}$$

or

Similarly, the probability of event B with known event A:

$$P(A \wedge B) = P(B \mid A)\, P(A) \qquad \text{(ii)}$$

Equating right-hand side of equations (i) and (ii), we get,

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)} \qquad \text{(iii)}$$

Equation (iii) is known as **Bayes' theorem.** It is fundamental to most modern AI systems for **probabilistic inference**. Here,

P(A | B) is **posterior probability** and will be read as probability of hypothesis A when an evidence B has occurred.

P(B | A) is the likelihood. If the hypothesis is true, we calculate the probability of evidence.

P(A) is the **prior probability**, that is, the probability of hypothesis before considering the evidence.

P(B) is called **marginal probability**, pure probability of an evidence.

We can also write equation (iii) as

$$P(A_i \mid B) = \frac{P(A_i) * P(B \mid A_i)}{\sum_{i=1}^{k} P(A_i) * P(B \mid A_i)}$$

where A_1, A_2, A_3,……., A_n is a set of mutually exclusive and exhaustive events.

Bayes' rule can also be used to perceive the effect of some unknown cause. In such a situation, Bayes' rule becomes:

$$P(\text{cause} \mid \text{effect}) = \frac{P(\text{effect} \mid \text{cause})\, P(\text{cause})}{P(\text{effect})}$$

Bayes' theorem is extensively used in AI to calculate the next step of the robot when the already executed step is given. It is also used in weather forecasting.

Example: What is the probability that a patient with a stiff neck is suffering from meningitis, given that

Meningitis causes a patient to have a stiff neck 80% of the time?

The known probability that a patient has meningitis is 1/30,000.

The known probability that a patient has a stiff neck is 2%.

Solution: Let A be the proposition that a patient has stiff neck.

Let B be the proposition that a patient has meningitis.

So, P(A | B) = 0.8.

P(B) = 1/30000.

P(A)= .02.

$$P(B \mid A) = \frac{P(A \mid B)\, P(B)}{P(A)} = \frac{0.8 * (1/3000)}{0.02} = 0.0013333$$

Hence, 1 patient out of 750 patients has meningitis disease with a stiff neck.

Example: From a standard deck of playing cards, a single card is drawn. The probability that the card is king is 4/52. Find out the posterior probability P(King|Face), which means the drawn face card is a king card.

Solution:

Let A be the probability that card A is a king.

Let B be the probability that card B is a face card.

P(B | A) be the probability of face card being a king.

We know that

P(A) = 4 / 52= 1 / 13

P(B) = 1/ 13

P(B | A) = 1

So,

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} = \frac{1 \star (1/13)}{3/13} = 1/3$$

Hence, probability that a face card is a king card = 1 / 3.

5.3.1 Bayesian Belief Network in Artificial Intelligence

In statistics, probabilistic models are used to define the relationship that exists between variables and the probabilities of each variable. However, in real-world problems, a number of variables is just too many. In such a scenario, a large amount of data is required to cover each case of probability functions. Calculating this data in real time is another big issue.

Therefore, we need a model that preserves the conditional dependencies between random variables and conditional independence in other cases. These requirements from a model are fulfilled by Bayesian networks, which help us to effectively visualize the probabilistic model for each domain and study the relationship between random variables in the form of a user-friendly graph.

Bayesian belief network, also known as a **Bayes network, belief network, decision network** or **Bayesian model**, is an important concept in computing that deals with probabilistic events to solve a problem which has uncertainty. We can define a Bayesian network as a probabilistic graphical model which represents a set of variables and their conditional dependencies using a directed acyclic graph.

The Bayesian network graph does not contain any cycles. Hence, it is known as a directed acyclic graph or DAG.

Bayesian networks are said to be probabilistic because they are created using probability distribution. Moreover, they use probability theory for prediction and anomaly detection. Since real-world applications are probabilistic in nature, we need a Bayesian network to represent the relationship between multiple events. For example, in various tasks including prediction, anomaly detection, diagnostics, automated insight, reasoning, time series prediction and decision-making under uncertainty, Bayesian networks are used extensively.

A Bayesian network is created using available data and experts' opinions. It is made of two parts as follows:

- Directed acyclic graph (DAG) (refer **Fig. 5.1**)
- Table of conditional probabilities

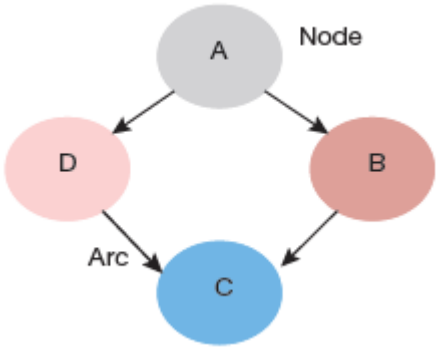


FIGURE 5.1 Directed acyclic graph (DAG)

A Bayesian network graph has multiple nodes and each node is connected to another node with an arc (or directed link). Here, the following points are noted.

1. Each **node** represents a random variable which can either be **continuous** or **discrete**.
2. **Arc or directed arrows** connect a pair of nodes in the graph and are used to depict the causal relationship between random variables.
3. An arc also indicates that one node is directly influencing the other node. In case there is no directed link between two nodes, then it simply means that nodes are independent with each other.

The generalized form of Bayesian network that represents and solves decision problems under uncertain knowledge is known as an **influence diagram**.

In **Fig. 5.1**, A, B, C and D are random variables represented by the nodes in the graph. If node B is connected to node A by a directed arrow, then node A is called the parent of node B. Note that node C is independent of node A.

- Each node in the Bayesian network has condition probability distribution $\mathbf{P(X_i \mid Parent(X_i))}$, which determines the effect of the parent on that node.
- Bayesian network is based on joint probability distribution and conditional probability.

5.3.2 Joint Probability Distribution

If we have variables x1, x2, x3,....., xn, then the probabilities of a different combination of x1, x2, x3.. xn, are known as joint probability distribution. That is, in terms of the joint probability distribution, $P[x_1, x_2, x_3, ..., x_n]$, can be written as,

$$= P[x_1 \mid x_2, x_3, ..., x_n]P[x_2, x_3, ..., x_n]$$
$$= P[x_1 \mid x_2, x_3, ..., x_n]P[x_2 \mid x_3, ..., x_n] ....P[x_{n-1} \mid x_n]P[x_n].$$

In general, for each variable Xi, the equation can be given as,

$$P(X_i \mid X_{i-1}, ..., X_1) = P(X_i \mid Parents(X_i)).$$

5.3.3 Understanding the Semantics of Bayesian Network

There are two ways to understand the semantics of the Bayesian network.

1. To understand the network as the representation of the joint probability distribution. This will be helpful to understand the construction of the network.
2. To understand the network as an encoding of a collection of conditional independence statements to design the inference procedure.

We have seen that a Bayesian belief network is a graphical representation of different probabilistic relationships among random variables in a particular set. It classifies without any dependency on attributes. This means that the classifier is condition-independent.

The probability in a Bayesian belief network is specified using the condition, P(attribute/parent), which gives the probability of an attribute, true over parent attribute.

Local Markov Property

Bayesian networks satisfy the Local Markov Property which states that a node is conditionally independent of its non-descendants (given its parents). For example, if P(D | A, B) is equal to P(D | A) because D is independent of its non-descendent, that is, node B. This property very useful in simplifying the joint distribution.

Example of Bayesian Networks

Assume that you need to model a student's marks (*m*) for an exam. Consider the Bayesian network graph given below that shows that marks depend upon two other variables—exam level and IQ level—which can be explained as,

exam Level (*e*) is a discrete variable which specifies the difficulty of the exam. Its value is either 0 (for easy) or 1 (for difficult).

The IQ Level represents the intelligence quotient level of a student. It is also a discrete variable and has a value 0 (for low) or 1 (for high). The IQ level of the student influences the aptitude Score of the student (***s***).

A student's admission to a particular university depends on the marks obtained and is given as follows:

In **Fig. 5.2**, conditional probabilities table or CPTs are given that represent the probability distribution values of the given variables. Note the following:

- The sum of the CPT values in each row is equal to 1.

- A Boolean variable having k parents, and $2^{\mathbf{K}}$ probability values. That is, if there is only 1 parent, then there will be $2^{\mathbf{1}}$ = 2 values in CPT. If there are 2 patents, then there are $2^{\mathbf{2}}$ = 4 values in CPT, and so on.

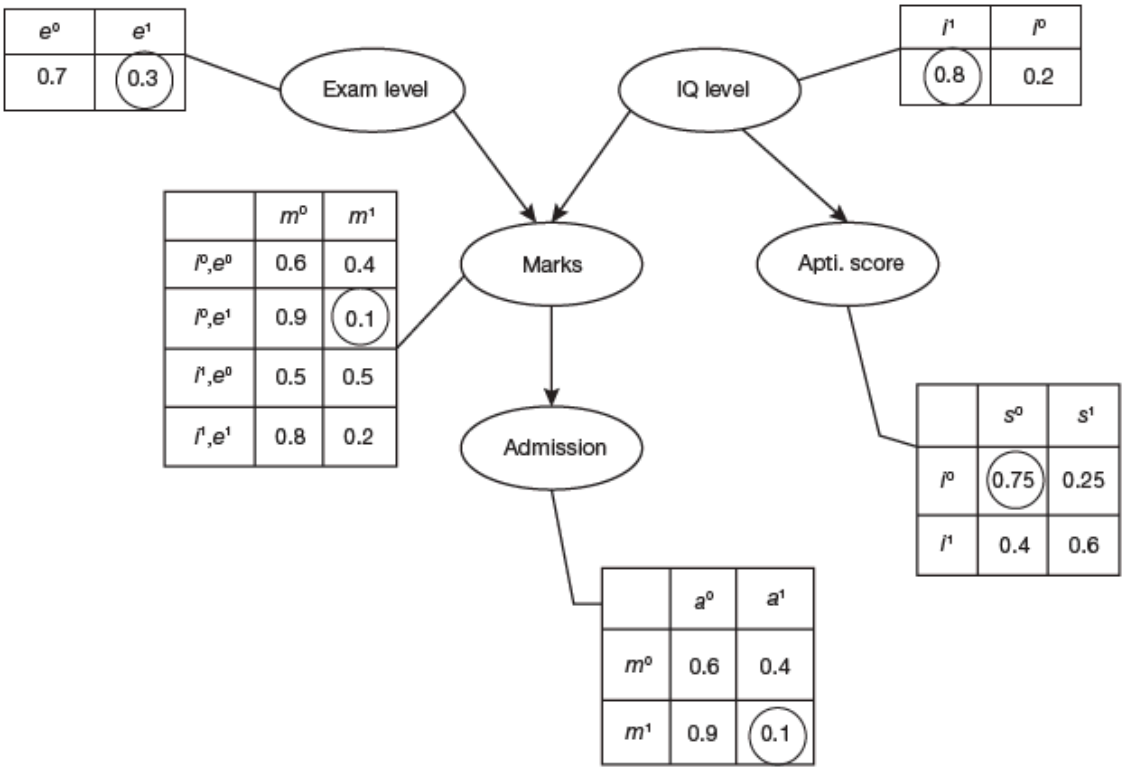


FIGURE 5.2 Conditional Probabilities for Admission Process

Let us first enumerate our five variables.

1. Exam level (e)
2. IQ level (i)
3. Aptitude score (s)
4. Marks (m)
5. Admission (a)

To calculate the joint probability distribution of the five variables, the formula is given by

$$P[a, m, i, e, s] = P(a \mid m) \cdot P(m \mid i, e) \cdot P(i) \cdot P(e) \cdot P(s \mid i)$$

which clearly indicates the following:

1. P(a | m) denotes the conditional probability of the student getting admission based on the marks scored in the examination.

2. P(m | i, e) represents the marks obtained by the student is determined by his/her IQ level and difficulty level of the exam.
3. P(i) is the probability of the IQ Level a.
4. P(e) is the probability of the exam level.
5. P(s | i) is the conditional probability of the student's aptitude score, given his IQ level.

Now, let us use the probabilities to calculate the joint probability distribution of the entire Bayesian network. We will consider two cases here.

Case 1: Calculate the probability when the exam level is difficult, the student has a low IQ level and a low aptitude score, but still manages to pass the exam and secure admission to the university.

Here, the joint probability distribution can be given as follows:

P[a=1, m=1, i=0, e=1, s=0]

Look at the conditional probability tables and note down the values for the given conditions.

Probability when a = 1 and m = 1 is 0.1,

m is dependent on i and e, and as per the question, we have to find the value given: m = 1, I = 0, e = 1. This probability = 0.1.

Since IQ level is low (i = 0), the value of probability for i is 0.8.

The exam level is difficult, so, e = 1. The probability for e = 1, will be 0.3.

The IQ level is low, thus score = 0. Hence, the value is 0.75.

Now, the joint probability can be given as

= 0.1 * 0.1 * 0.8 * 0.3 * 0.75 = **0.0018.**

Case 2: Calculate the probability that a student has high IQ level and aptitude score, the exam was easy but could not manage to pass the exam.

The formula for the JPD in this case is

P[a=0, m=0, i=1, e=0, s=1].

Looking at the tables, we get

No admission because of less marks (a=0, m =0) = 0.6

Less marks despite high IQ and easy exam (i=1, e=0) = 0.5

Exam easy = 0.7 High IQ (i=1) = 0.2 Good score (s=1) = 0.6

= 0.6 * 0.5 * 0.2 * 0.7 * 0.6

= 0.0252.

5.4 Learning

Herbert Simon defined learning as changes in a system that enable a system to do the same task more efficiently the next time.

To understand the term ‘learning’ in the context of computers, let us go back to 1997 when Mitchell proposed, ‘A computer program is said to learn from experience **'E'** with respect to some class of tasks **'T'** and performance measure **'P'**, if its performance at tasks in **'T'**, as measured by **'P'**, improves with experience **E** ‘.

Taking the concept of learning in computers to the next level, machine learning, is defined as a branch of computer science that is specially dedicated to study and design algorithms that can produce predicates from the given dataset.

Machine learning facilitates computers to learn without being explicitly programmed. The agents learn not only to perform a particular task but also to improve their future performance.

The following tasks must be learned by a machine learning agent.

1. Predict the result state for an action.

2. Know the values for each state and interpret which state has high or low value.
3. Record all relevant precepts.

Why do we require machine learning?

1. Machine learning helps in improving and understanding the efficiency of human learning.
2. Machine learning is used to discover new things that were otherwise not known to us.

5.4.1 Components of a Learning System

The main components of a learning system include the following.

1. ***Performance Element:*** It is the agent that acts in the world. The action is performed by interpreting (or precepting) events in real world and then deciding what actions need to be taken.
2. ***Learning Element:*** The task of this element is to make improvements in the system by accepting knowledge about the performance element and providing feedback to determine ways to make changes in the performance element.
3. ***Critic:*** As the name suggests, this element tells the learning element how the agent is doing. For this, it compares the agent's performance with the fixed standard of performance.
4. ***Problem Generator:*** This component suggests problems or actions that will generate new examples or experiences. This additional knowledge would help the system to further train itself.
   For example: Self-driving cars

5. ***Performance Element:*** It contains knowledge about procedures for driving. For example, knowledge about turning, applying breaks and accelerating.
6. ***Learning Element:*** It formulates goals. For example, the learning element learns the procedure to apply breaks, accelerate the car, take a safe turn, learn the geography of the city, etc.
7. ***Critic:*** It observes real-world scenario and communicates the information to the learning element. For example, the element observes the

reaction of other drivers when a sharp turn is taken by the car.

8. ***Problem Generator:*** It generates a new problem like 'try the congested north city road'.

5.4.2 Forms of Learnings

There are different types of learnings which are discussed below.

5.4.2.1 Rote Learning

Rote learning has the following features.

1. It is based on memorization.
2. It ignores the inner complexities.
3. The learner just recalls the stored knowledge.

For example, human learners learn a poem by repeatedly reciting it without understanding its actual meaning.

5.4.2.2 Induction Learning

Induction learning has the following features.

1. It is also known as learning by example.
2. It is based on supervised learning.
3. In the learning process, a general rule is induced by the system from a set of observed instances.
4. Class definitions are constructed using any classification technique.

For example, if '***f***' is a function of input x, then f(x) denotes the output function applied to 'x'.

To find the hypothesis h such that h ≈ *f*, we can plot three graphs. Consider **Fig. 5.3** (a) that shows points (x,y) given in a plane so that y = *f*(x). Our job is to find a function h(x) that fits the point well.

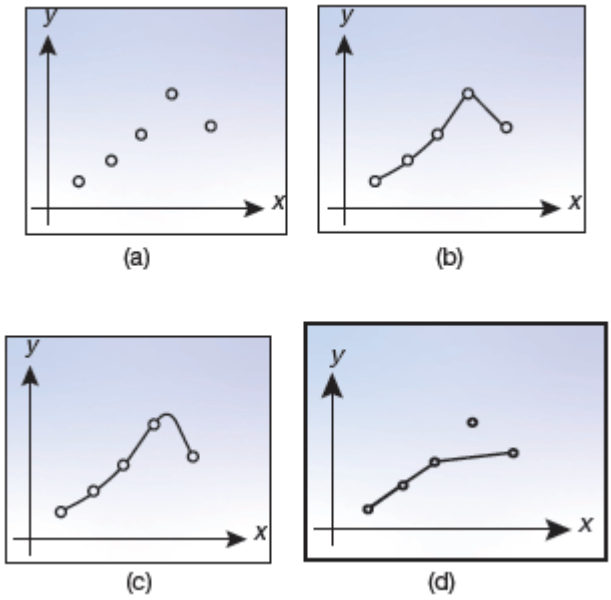


FIGURE 5.3 Plotting graphs for $y = f(x)$ to find a solution that best fits the points

Note that in **Fig. 5.3 (b)**, a linear function 'h' is connecting the points. In **Fig. 5.3 (c)**, a more complicated function is used. So, even though input points are the same, the output is different. **Figure 5.3 (d)** is different from the graphs given in **Figs 5.3 (b)** and **5.3 (c)** as it completely points and fits others with a simple function.

5.4.2.3 Learning By Taking Advice

This type of learning has the following features.

1. It is the easiest way of learning.
2. It follows a simple technique.
3. A programmer writes a program to give some instructions to perform a task. Once it is programmed or learned, the system will perform its tasks.
4. We human learners take advice from several sources including other humans (experts), the Internet, etc.
5. This type of learning works by deriving inference rather than just doing rote learning.
6. With more and more stored knowledge getting transformed into an operational form, the reliability of the knowledge source improves.

5.4.2.4 Explanation-Based Learning (EBL)

This type of learning has the following features.

1. It deals with an idea of single-example learning.
2. It requires a substantial number of training instances.
3. The main issue is that having a number of training instances is difficult.
4. It helps to learn certain things effectively, specially when sufficient knowledge exists.
5. While instance-based learning is data-driven and thus data-intensive, EBL, on the other hand, is more knowledge-intensive and knowledge-driven.
6. The process of EBL can be explained in the following manner:
    1. Initially, an EBL system accepts a training example.
    2. The operationality criteria and domain theory are analysed. This includes a set of rules that describe relationships between objects and actions in a domain.
    3. The training example is then generalized to describe the goal concept and to satisfy the operationality criteria.
    4. Thus, several applications are possible for knowledge acquisition and engineering aspects.

5.4.2.5 Reinforcement Learning (Problem Solving)

We have already seen in **Chapter 1** that reinforcement learning is a feedback-based machine learning algorithm in which an agent learns to act in a restricted environment. For this, it performs certain actions and then observes the results of those actions. For each correct action, the agent gets a reward and for each bad action, the agent gets negative feedback or a penalty.

Although the program designer sets the reward policy, no hint or suggestion is given to the learner about how to solve the task at hand. The learner has to figure out how to perform the task in such a way that as to get the maximum reward. For this, it starts from totally random trials and ends with sophisticated tactics and superhuman skills.

Reinforcement learning is the best way to enhance a machine's creativity. When run on a powerful parallel computer infrastructure, machines can be made to learn from the experiences of thousands of other learners learning parallelly. Let us quickly summarize the concept of problem solving before going further.

1. Humans prefer to learn by solving various real-world problems.
2. Reinforcement learning is based on this form of solving real-world problems.
3. We know that repeating certain actions results in a desirable outcome but avoiding those actions can lead to undesirable outcomes.
4. Therefore, we need to evaluate outcomes using a utility function. The utility function is the worth of a particular outcome.
5. There are several research issues which include the identification of the learning rate, time and complexity of the algorithm, convergence, representation (frame and qualification problems), handling of uncertainty (ramification problem), adaptivity and 'unlearning', etc.

In reinforcement learning, the system knows the desirable outcomes but does not know which actions should be performed to get those outcomes.

In such a case, the effects of performing the actions are usually compounded with side-effects. This makes it impossible to specify the actions to be performed in accordance to the given parameters.

Q-Learning is the most widely used reinforcement learning algorithm. It uses a simple value iteration update technique. For each state 'S', from the state set S, and for each action, a, from the action set 'A', it is possible to calculate an update to its expected reduction reward value, with the following expression:

$$Q(s_t,a_t) \leftarrow Q(s_t,a_t) + a_t(s_t,a_t)[r_t + \gamma \max_a Q(s_{t+1},a) - Q(s_t,a_t)]$$

where

r_t is a real reward at time t,

$\alpha_t(s,a)$ are the learning rates such that $0 \leq \alpha_t(s,a) \leq 1$ and

γ is the discount factor such that $0 \leq \gamma < 1$.

5.2.4.6 Rote Learning

We all know that rote learning completely ignores the underlying complexities in the subject. It just focuses on memorizing the content so that the learner can recall it exactly in the same way it was learnt (read or heard).

Learning can be done either by memorization or repetition. In memorization, the learner just retains the concepts learnt without understanding them. In repetition, the learner says the same thing again and again to remember it by heart. For example, a child recites a poem multiple times to retain it.

5.2.4.7 Chunking Memory

Chunking is more efficient than normal memorizing. Here, chunking refers to the organization of information. Chunking is the process of encoding memories into long-term memory for later recall. It helps to get above the problem of short-term memory. According to a research study by George Miller in 1956, a person can handle between five and nine pieces of information at the same time.

Miller also introduced the term ‘chunking’ to help people retain more pieces of information.

For example, let us look at our phone number, 001 – 725-258-8901. This American phone number is quite long. To memorize this number, we can break it in chunks and then learn those chunks by associating a meaning with it. Here, 001 is the country code, 725 is the area code and the last seven digits is the actual number.

Teachers and parents also use chunking to help students memorize information in their long-term memory. In our daily lives, we use chunking; for example, to recognize a person’s face, we use several stored pieces of information about that face.

A related concept called cognitive loading theory states that the amount of information and interactions that must be processed simultaneously can either underload or overload the finite working memory. If the memory is overloaded, then all the elements must be processed to learn the entire concept. More the information a person has to learn in a shorter period of time, higher is the difficulty to process that information.

Richard E. Mayer, an educational psychologist, has stated that people learn better when a complex continuous lesson is broken into separate segments. For example, a complex figure can be broken into two or more smaller figures. Each smaller figure should present a different part of the original one. In this way, the learner's working memory is less likely to be overloaded with essential processing.

Jerome Bruner proposed the concept of spiral learning which refers to the idea of revisiting basic ideas. Each time an idea is revisited, it is elaborated further.

5.2.4.8 Inductive Learning

There are two types of inductive learning—supervised and unsupervised.

Supervised Learning

In this type of learning, the machine has access to a teacher who corrects it. The machine learning algorithm uses an inference function to train it using labelled training data. The training data consist of an input-output pair. For example, if we input the image of a fruit, the machine learning algorithm must be able to identify whether that fruit is an apple or mango.

Unsupervised Learning

Contrast to supervised learning, unsupervised learning does not have any access to the teacher. Instead, the machine must itself search for 'order' and 'structure' in the environment. There is no desired output in this learning technique. Therefore, categorization is done so that the algorithm correctly differentiates between two (or even more) different entities. Unsupervised learning is usually done when there is no labelled

data. Here, we do not know which entity belongs to which category. For example, we do not know whether a person is suffering from COVID or not. We just see certain symptoms and then categorize that the person is COVID+. Similarly, the machine does not know whether a person will buy a new product or not. The machine learning algorithm just accepts data about the new customer and then categorizes him/her as a potential buyer by comparing his/her details with those who had bought that product in the past. If the new customer's data seem to be similar with existing customers, then he/she is a potential buyer; else not.

5.5 Clustering

Clustering is the process of grouping together objects in a single cluster that have similar properties. Objects in a cluster are similar to each other and objects in different clusters vary significantly from each other.

There are different algorithms for clustering data. For example, hierarchical clustering specifies the hierarchy of clusters and depicts the relationship that exists between two clusters. Distance-based clusters can be used to find out similarities between two data points as closer data points are more similar than those further away. Similarly, density-based clustering defines clusters based on the density of data sets and centroid-based clustering forms clusters by finding the nearest cluster centre (which may not be a data point).

Clustering is better than classification as it is adaptable to changes and identifies features that distinguish different groups.

Irrespective of the technique used for clustering, labels are assigned to each cluster. Note that clusters can vary depending on the clustering algorithm used. So, it is always recommended to perform multiple analyses by running different clustering algorithms on the same data to get better results.

Clustering, which is an example of unsupervised learning, is used when the target features are not given in the training examples. The technique

constructs clusters which can then be used to predict feature values for the examples in the class.

Different clustering algorithms give different prediction errors. But, at the end, the best clustering is technique, that minimizes the error, is chosen.

Example of Clustering: In college, your teacher may want to cluster students' learning behaviour so that he/she can implement different strategies to deal with different sets of students.

5.5.1 Clusters

Clustering is a set of techniques used to partition data into a number of groups, also called clusters. A c**l**uster can be considered as a group of data objects that are similar to other objects in the same cluster and dissimilar to data objects in other clusters. Clustering helps data scientists identify two main qualities of data—meaningfulness and usefulness.

Meaningful clusters enhance domain knowledge. For example, in the healthcare industry, clustering is used to make groups of patients whose bodies responded differently to the same medical treatment.

Useful clusters serve as an intermediate step in a data pipeline. For example, businesses use this technique for making different clusters of customers depending on their purchasing patterns to create targeted advertising campaigns.

5.5.2 Overview of Clustering Techniques

Data objects can be clustered using various techniques. Each of these techniques has its own strengths and weaknesses. Depending upon the input data, we need to select the most appropriate technique that results in more natural cluster assignment. To select the most suitable clustering algorithm for the dataset is not a trivial task. Some important factors that affect this decision include the characteristics of the clusters, the features of the dataset, the number of outliers and the number of data objects.

Based on these features, three popular categories of clustering algorithms include the following:

1. Partitional clustering
2. Hierarchical clustering
3. Density-based clustering

Partitional Clustering

It divides data objects into non-overlapping groups (refer **Fig. 5.4**) in such a way that every object belongs to one and only one cluster, and every cluster must have at least one object.

In such algorithms, users can specify the number of clusters (indicated by the variable *k)*. These algorithms iteratively assign subsets of data points into *k* clusters. Two examples of partitional clustering algorithms are *k*-means and *k*-medoids. Note that these algorithms are **nondeterministic.** That is, such algorithms produce different results from two separate runs even if the same data was used in every run.

Advantages

1. It works well when clusters have a **spherical shape**.
2. **It is scalable** with respect to algorithm complexity.

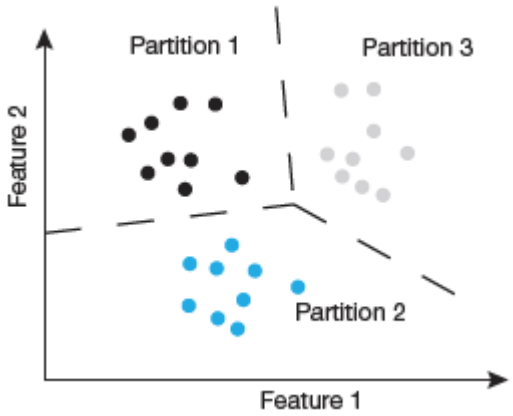


FIGURE 5.4 Partitional Clustering

Limitations

1. It does not perform well with **complex cluster shapes** having different sizes.
2. It does not work with clusters of different **densities**.

Hierarchical Clustering

It creates clusters in a hierarchy either by using a bottom-up or a top-down approach.

In agglomerative clustering, a bottom-up approach is used to merge two points that are the most similar. This process is repeated until all points have been merged into a single cluster.

In divisive clustering, a top-down approach is used in which the algorithm starts with all points as one cluster (refer **Fig. 5.5**). In each iteration, the data objects that are the least similar in the cluster are split until only single data points remain.

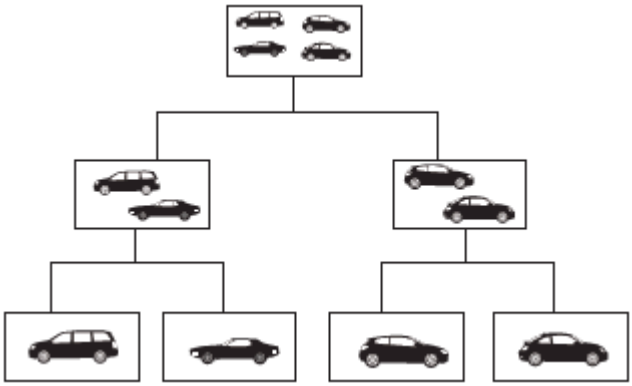


FIGURE 5.5 Divisive Clustering

In such an algorithm, a tree-based hierarchy of points called a **dendrogram** is created. Number of clusters to be created, that is, k is determined by the user. Clusters are created by cutting the dendrogram at a specified depth that results in *k* groups of smaller dendrograms.

Hierarchical clustering is a **deterministic** process. Therefore, cluster assignments do not change when the algorithm is executed two or more times using the same input data.

Advantages

1. They highlight the **relationships** between data objects at a finer level.
2. Results are easily **interpretable.**

Limitations

1. They're **computationally expensive** with respect to algorithm complexity.
2. They are adversely affected by **noise** and **outliers** in the data set.

Density-Based Clustering

This technique creates clusters based on the density of data points in a region. A cluster is created where there are high densities of data points separated by low-density regions (refer **Fig. 5.6**).

In contrast to other clustering categories, density-based approach does not require the user to specify the number of clusters. It uses a distance-based parameter that acts as a threshold. The threshold value helps the algorithm to determine how close points must be to be considered a cluster member.

Density-Based Spatial Clustering of Applications with Noise (or DBSCAN) and Ordering Points to Identify the Clustering Structure (or **OPTICS**) are popular examples of density-based clustering algorithms.

Advantages

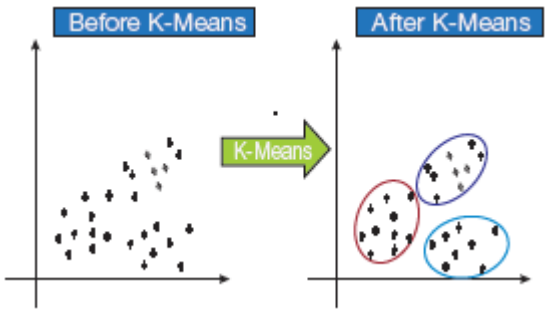


FIGURE 5.6 Density-Based Clustering

1. It works well when clusters are of **non-spherical shapes**.
2. It performs well even on a data set having noise or outliers.

Limitations

1. It is not well suited for clustering in **high-dimensional spaces**.
2. It is difficult to identify clusters of **varying densities**.

5.6 Explanation-Based Learning (EBL)

Explanation-based learning learns from a single training instance. With that instance and a functional description, EBL builds a general structural description (refer **Fig. 5.7**). An EBL system accepts four kinds of input described as follows:

1. A training example which states what the learning *sees* in the world.
2. A goal concept is a high-level description that states what the program is supposed to learn.
3. An operational criterion specifies a description of concepts that are usable. It states which other terms appear in the generalized result.
4. Domain theory contains a set of rules that describe relationships between objects and actions in a domain.

With these four inputs, the EBL computes a generalization of the training example that describes while satisfying the operational criterion. This process is completed in two steps.

Step 1: Explanation: The domain theory is used to prune all irrelevant aspects (with respect to the goal concept) of the training example.

Step 2: Generalization: The explanation obtained in Step 1 is generalized as far as possible in such a way that it still describes the goal concept. Generalization ensures that all cases are covered and not just specific ones.

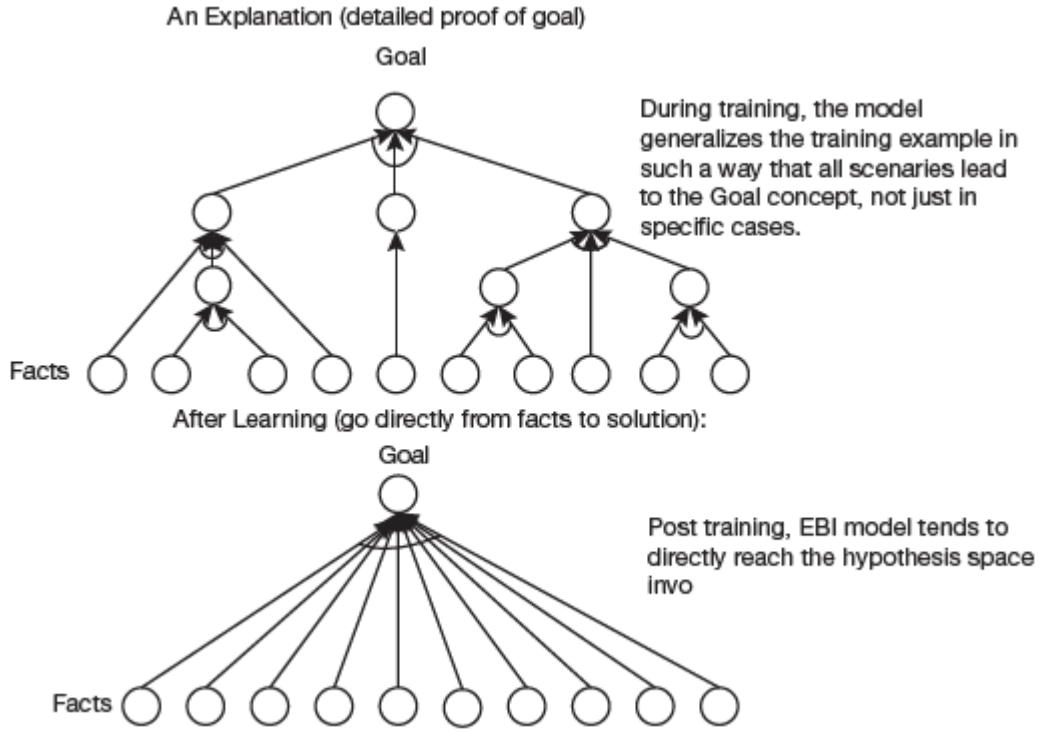


FIGURE 5.7 Explanation-Based Learning

Therefore, we see that EBL uses an explicit **domain theory** to generate an **explanation** or **proof** of a training example. The proof is then generalized and the new knowledge acquired is then applicable to even non-training situations.

EBL differs from **inductive** learning as new knowledge is added in the domain theory. Therefore, EBL is sometimes called **deductive learning** or **analytic learning**.

Example: Learning that an Object is a Mug

Target Concept: mug(C) :- premise(C).

Domain Theory:

```
mug(X) :- liftable(X),
     holds_liquid(X).
holds_liquid(Z) :- component (Z,W),
     OpenCylinder(W),
     points_up(W).
liftable(X) :- less_weight(X), component(Y,handle).
less_weight(X) :- small(X).
```

We will use knowledge specified in the domain theory to determine when some object is called a mug. We want to derive an explicit rule that specifies when something is identified as a mug.

Training Example: An Instance of a mug

```
mug(obj1) small(obj1) component(obj1,handle)
has(Krish,obj1) component(obj1,bottom)
     component (obj1, OpenCylinder)
points_up(OpenCylinder)
looks_like(OpenCylinder)
color(obj1,yellow)
```

Proof (or Explanation): First, we will prove that obj1 is a mug as shown in Fig. 5.8.

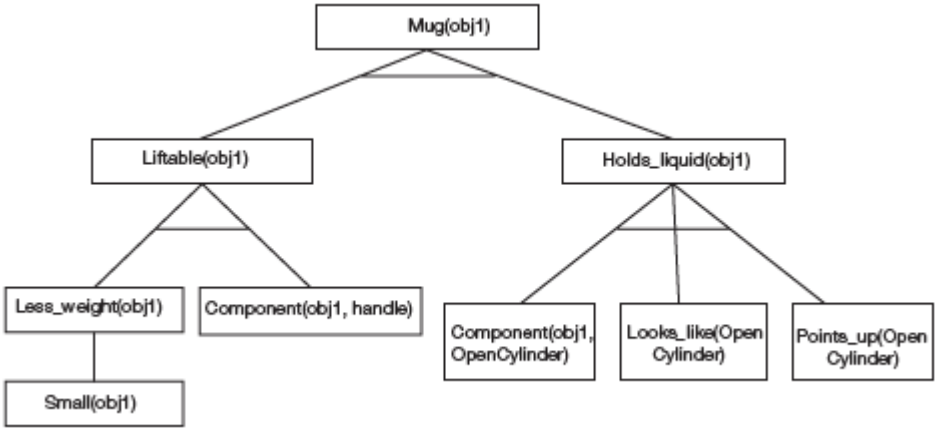


FIGURE 5.8 Proof of obj1 is a mug.

Next: Generalize the proof to: X is a mug. In this step, we generalize all constants that depend on the training example. In our training example (refer **Fig. 5.9**), OpenCylinder and **obj1** are the constants.

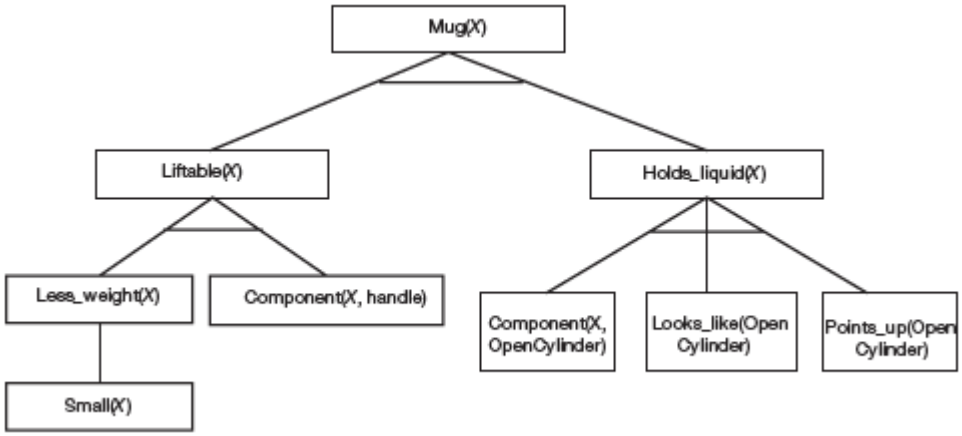


FIGURE 5.9 Generalize the proof to: X is a mug

Next: Add the new *chunk of knowledge* to the domain theory.

```
mug(X) :- small(X),
    component(X,W),
    looks_like(W),
    points_up(W).
```

'Pure' EBL only learns how to solve faster, not how to solve problems previously insoluble.

Note that we have ignored all irrelevant information in the training example while formulating the proof or adding new knowledge to the domain theory. Once added, the new rule(s) can be used to identify any similar object as a mug.

Thus, the EBL agent learns by examining particular situations and relating them to gained knowledge base. The agent uses this gained knowledge to solve similar types of problems. Let us now try to understand the architecture of EBL shown in **Figs 5.10** and **5.11**.

The **problem solver** takes inputs from the environment. These inputs include a training example, inference rules and the goal concept. The problem solver processes these inputs and gives justification to generalizer. This consists of the following areas:

Goal concept is the problem statement stated in a complex form. The EBL agent needs to learn it.

Training examples are facts which explain an instance of the goal concept.

Inference rules represent the facts that the learner already knows.

The generalizer accepts the output of the problem solver as its input. Its job is to compare the explanation of the problem solver with the knowledge base. The output of this comparison is passed to the operationality pruner.

The operationality pruner accepts the output of the generalizer and the operationality standard as inputs. The operationality standard describes the final concept and specifies the form in which learned concept should be expressed.

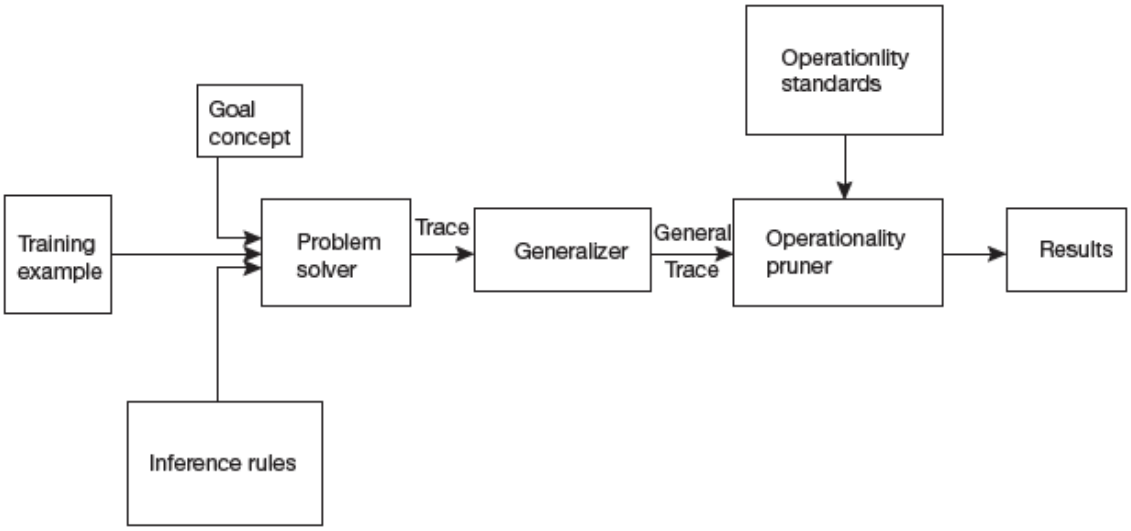


FIGURE 5.10 Architecture of EBL

Example: Consider the domain theory and the facts given below to make the EBL learner to identify that Robot(r) ^ R2D2(r) → Robust(r)

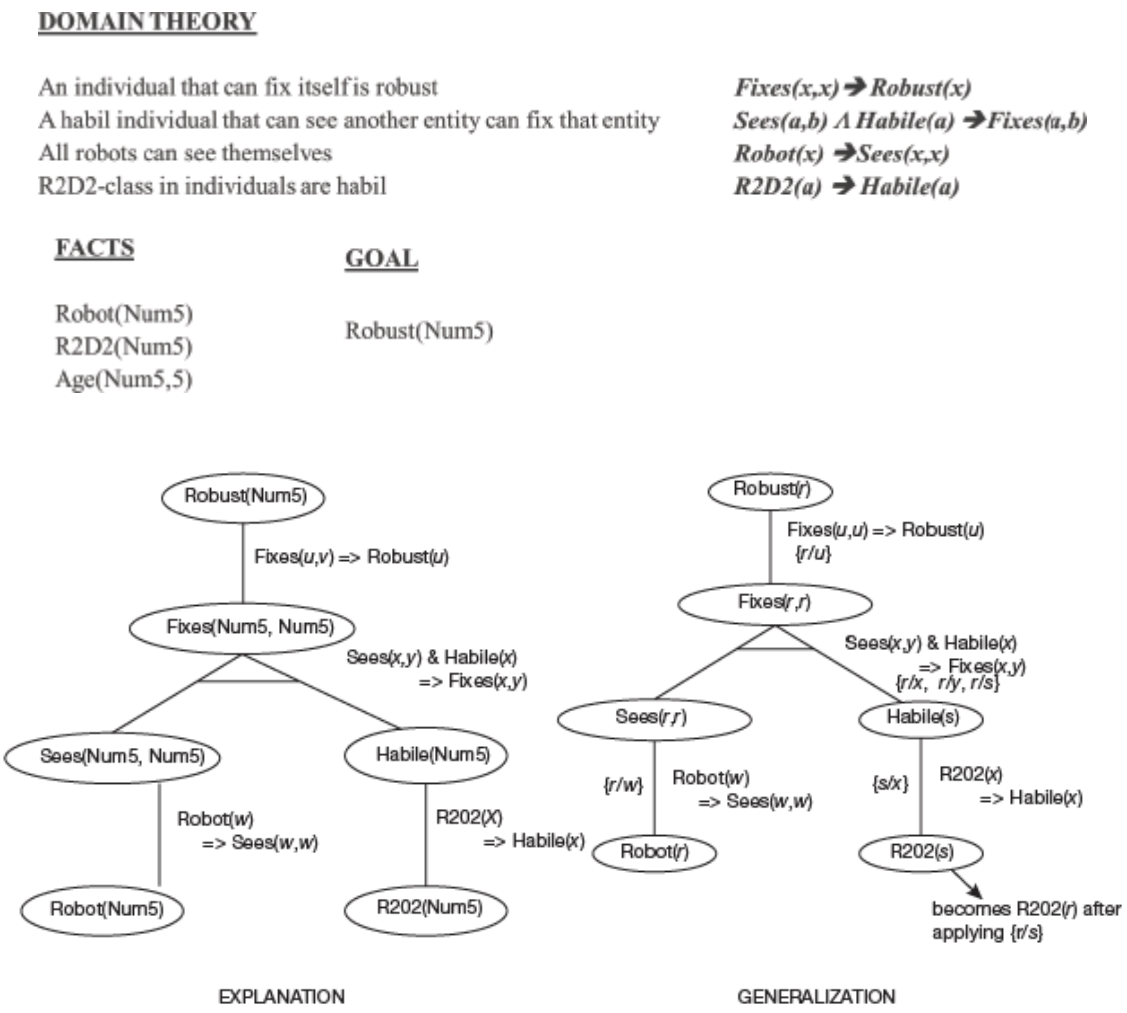


FIGURE 5.11 Architecture of EBL based on given Domain Theory

5.7 Reinforcement Learning

Reinforcement learning is an area of machine learning that facilitates machines to find the best possible behaviour or path that should be taken in a specific situation. Reinforcement learning is different from supervised learning. While in supervised learning, the training data has labels that are used to train the models, reinforcement learning, on the other hand, is used when no labels exist. The agent decides actions to be performed to complete the given task. In the absence of a training dataset, it is bound to learn from its experience.

Example: The agent wants to take the reward, but there are many hurdles in between. The agent has to find the best possible path to reach the reward. In **Fig. 5.12**, note that our agent is a robot, the reward is a diamond and the hurdle is fire. The robot learns by trying all the possible

paths and then selects the one which gives it the reward with the least hurdles.

Each step correctly taken will get the robot, a reward. Correspondingly, a wrong step will subtract the reward of the robot. The total reward is calculated when the robot reaches the diamond.

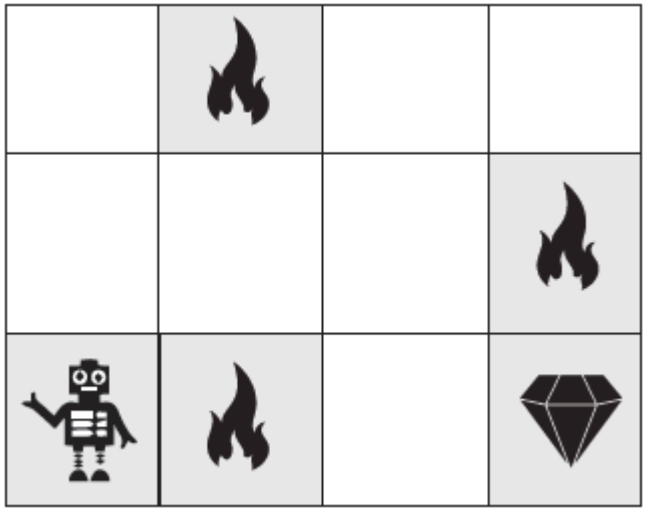


FIGURE 5.12 The Agent and the Reward

5.7.1 Components of Reinforcement Learning

A reinforcement learning system contains the following components.

1. ***Input:*** The input of a reinforcement learning system is an initial state from which the agent starts.
2. ***Output:*** Since there can be different solutions to a particular problem, a reinforcement learning system can have many possible outputs.
3. ***Training:*** Training depends on input. The agent returns a state and the user observes the output to decide whether the agent should get a reward or be punished.

Apart from these, we have the following components as well (as shown in **Fig. 5.13**):

1. ***Agent or the learner:*** An entity which performs actions in an environment to gain some reward is called an agent. The agent perceives

and explores the environment before acting upon it.



FIGURE 5.13 Components of Reinforcement Learning

2. ***Action:*** They are the moves taken by an agent in the environment.
3. ***Environment:*** This is the scenario in which the agent interacts and performs actions. The environment is assumed to be random in nature.
4. ***Reward (R):*** An immediate return given to the agent on performing a correct action or task is a reward. It acts as feedback to evaluate the action of the agent.
5. ***State (s):*** It refers to the current situation of the environment after an action is taken by the agent.
6. ***Policy (π):*** It is a strategy applied by the agent in the current state to decide the next action.
7. ***Value (V):*** It specifies long-term return with discount. It does not cater to short-term reward.
8. ***Value Function:*** **It is t**he function which returns the value of a state. This value returns the total amount of reward. The agent is expected to start from this state.
9. ***Model of the environment:*** **It is a** model that mimics the actual behaviour of the environment. It helps to draw inferences as it helps to observe how the environment will behave in a particular scenario.

The agent continues to learn until it finds the best solution that gives the maximum reward.

5.7.2 Example: Reinforcement Learning in Real World

When you bring home a pet, you train it to perform actions in an acceptable manner. How do you do that?

Since our pet does not understand our language, we cannot directly instruct what has to be done and how it has to be done. So, we emulate a situation in which the pet has to act in different ways. If the pet responds in the desired way, we give it a piece of flesh as the reward, else frown at it to make it understand that the act was undesirable.

After receiving the reward or the stern looks, the pet learns that if it performs a particular set of actions again in future in a similar type of environment, then it will get a reward. So, the pet performs more enthusiastically the next time by learning from positive experiences. At the same time, it also learns what not to do when faced with negative experiences.

Here, the following can be noted:

1. The pet is an agent that interacts and acts in the environment.
2. The environment is your house.
3. An example of a state could be the pet sitting in a corner and you using a specific word for the pet to walk.
4. The agent reacts by performing an action. This results in transition from one state to another. For example, the pet goes from sitting to walking.
5. The policy is a method of selecting an action.
6. After the transition, the pet may get a reward or penalty in return.

5.7.3 Characteristics of Reinforcement Learning

1. There is no supervisor. Only rewards direct the actions to achieve outcomes.
2. It involves sequential decision making
3. It is based on the hit and trial method.
4. Time is an important factor in reinforcement problems.
5. The agent performs action and changes states based on the feedback of the previous action.

6. Feedback is often delayed and is not available right at that instant.
7. What data the agent will receive depends on the agent's actions.
8. The environment is stochastic (random), so the agent needs to explore it to get the maximum positive rewards.

5.7.4 Differences Between Reinforcement Learning and Supervised Learning

The differences between reinforcement learning and supervised learning are given in **Table 5.1**.

TABLE 5.1 Differences between Reinforcement learning and Supervised learning

Reinforcement Learning	Supervised Learning
Output depends on the state of the current input. This means that the next input depends on the output of the previous input.	The output is made on the initial input or the input given at the start
Since decision is dependent, labels are given to sequences of dependent decisions	Decisions are independent of each other so labels are given to each decision.
Works while interacting with the environment.	Works based on examples or given sample data.
Works better where human interaction is prevalent.	Usually operated with an interactive software system or applications.
Example: Chess game	Example: Object recognition

5.7.5 Types of Reinforcement

Reinforcement learning (RL) is the science of making decisions and learning the optimal behaviour in an environment. This optimal behaviour is learned by continuously interacting with the environment and observing how it responds. The process of discovering an optimal solution is like a trial-and-error search. There are two types of reinforcement—positive and negative.

Positive Reinforcement

A**n** event that occurs due to a particular behaviour and increases the strength and the frequency of the behaviour, thereby causing a positive

effect on it is known as positive reinforcement.

Advantages

1. Maximizes performance
2. Sustains change for a long period of time

Disadvantage

Too much reinforcement may result in overloaded states which can diminish the results.

Negative Reinforcement

An event that occurs due to a particular behaviour and increases the strength of the behaviour by stopping or avoiding a negative condition is known as negative reinforcement.

Advantages

1. Enhances behaviour
2. Helps you define the minimum standard of performance
3. Gives performance better than the minimum standard of performance
4. It just provides enough to meet up the minimum behaviour

Reinforcement learning algorithms can also be categorized as model-free and model-based.

Model-Free Algorithms

These algorithms do not build an explicit model of the environment. They are based on trial-and-error algorithms that run experiments with the environment using actions to directly derive optimal policy. Model-free algorithms are either value-based or policy-based.

Value-Based Algorithms

Value-based algorithms formulate optimal policy by estimating the value function of every state accurately. The aim is to maximize a value function **V(s)** as the agent is expecting a long-term return of the current

states under policy π. The value function depends on the reward as, without reward, there could be no value. Thus, the values are estimated to achieve more rewards.

Some popular value-based algorithms are SARSA and Q-learning.

Policy-Based Algorithms

Policy-based algorithms compute the optimal policy without modelling the value function. It parametrizes the policy using learnable weights to render the learning problem into an explicit optimization problem. Popular policy-based Reinforcement Learning algorithms include Monte Carlo policy gradient (REINFORCE) and deterministic policy gradient (DPG). These approaches suffer from a high variance which results in instability during the training process.

Policy defines how an agent behaves at a given time. It maps the perceived states of the environment to the actions taken on those states.

In a policy-based RL method, the aim is to formulate a policy in such a way that the action performed in every state helps to gain maximum reward in the future. Two types of policy-based methods are as follows:

1. ***Deterministic:*** For any state, the same action is produced by the policy π.
2. ***Stochastic:*** Every action has a certain probability. And this probability will determine the action.

Hybrid algorithms combine value-based and policy-based approaches. Value-based approaches are more stable but are not suitable to model continuous action spaces. To gain the advantages of both the approaches, hybrid algorithms preferred. One of the most powerful RL algorithms—the actor-critic algorithm—is an example of a hybrid algorithm. In this algorithm, both the policy (actor) and the value function (critic) are parametrized to effectively use training data along with stable convergence.

Model-based reinforcement learning, however, focuses on the creation of a virtual model for each environment. This helps the agent to learn to perform in that specific environment. Note that there is no particular solution or algorithm for this approach because the model representation is different for each environment.

With the help of a model, inferences can be made to know how the environment will behave. For example, given a state and an action, a model can predict the next state and reward.

The model is usually used for planning as it facilitates taking up a series of actions by considering all future situations before actually experiencing those situations.

5.7.6 Applications of Reinforcement Learning

Applications of reinforcement learning include the following:

1. Robotics for industrial automation.
2. Aircraft control and robot motion control
3. Planning a business strategy
4. Machine learning and data processing
5. Design training systems, providing customized instruction and materials according to student's requirements

5.7.7 When to Use Reinforcement Learning in a Large Environment?

Reinforcement learning can be used in large environments in the following situations:

1. The environment is known, but an analytic solution is not known
2. Only a simulated environment is available.
3. The only way to collect information about the environment is to interact with it.

5.7.8 Benefits of Reinforcement Learning

Reinforcement learning is applicable to a wide range of complex problems that cannot be handled with other machine learning algorithms.

Reinforcement learning algorithms **seek a long-term goal** while exploring different possibilities autonomously, so they are said to be closer to artificial general intelligence (AGI).

Reinforcement learning algorithms focus on the problem as a whole. Unlike traditional machine learning algorithms that excel at specific subtasks, RL algorithms see the big picture. They do not divide the problem into subproblems. Rather, their goal is to maximize long-term rewards and for this, they may trade-off short-term rewards for long-term benefits.

RL algorithms **do not need a separate data collection step.** They obtain data through direct interaction of the agent with the environment. So, training data is available through the learning agent's experience and no additional data has to be fed to the algorithm.

RL algorithms work well in dynamic and uncertain environments as they are inherently adaptive and can quickly respond to changes in the environment. Unlike traditional machine learning algorithms, time and agent's experience are important factors in RL algorithms.

5.7.9 Challenges with Reinforcement Learning

Although RL algorithms can successfully solve complex problems in diverse simulated environments, their adoption in the real world has been slow due to the following reasons.

Needs Extensive Experience to Train the Agent

The data is collected when the agent interacts with the environment. Therefore, the rate of data collection is limited by the dynamics of the environment. Environments with high latency slow down the learning curve.

Finding a good solution requires extensive learning and exploration in complex environments with high-dimensional state spaces.

The learning agent has to trade off short-term rewards for long-term gains. But in situations where the outcome is unknown, until a large num-

ber of sequential actions are taken, it may be difficult for the agent to discover the optimal policy. For example, in a game of chess, the outcome is unknown until both players have made all their moves.

RL Algorithms Lack Interpretability

We know that an agent once trained to work with optimal policy and deployed in the environment, takes actions based on its experience. To an external observer, the reason for these actions might not be clear. This lack of interpretability hampers the trust between the agent and the observer. If an observer can explain the actions taken by the agent, then the problem can be understood in a better way and the limitations of the model can be easily discovered, especially in high-risk environments.

Other miscellaneous challenges include the following:

1. Parameters may affect the speed of learning.
2. Real-world environment may have partial observability and may be constantly changing.
3. Too much reinforcement at times results in overload of states that affects the results.

5.7.10 Future of Reinforcement Learning

Traditionally, reinforcement learning algorithms could perform only one task at a time and each task is learned by a separate agent. These agents do not share knowledge. This makes learning complex behaviour (like driving a car) inefficient and slow.

Since certain problems share a common information source, have related underlying structure, and are interdependent on each other, they can get a huge performance boost if multiple agents work together and share their information with each other. For example, A3C or Asynchronous Advantage Actor-Critic is an algorithm in which related tasks are learned concurrently by multiple agents. Such multi-task learning scenarios will ultimately drive reinforcement learning closer to AGI.

Moreover, the trend is towards deep reinforcement learning that uses deep neural networks to model the value function (value-based) or the agent's policy (policy-based) or both (actor-critic). With deep neural networks, relevant features are automatically generated during the training process, thereby allowing the agent to learn optimal policies in complex environments.

While in RL, complex features had to be engineered to train the algorithm, there is no such limitation in deep RL. So, reduced learning capacity, limiting the scope of RL to simple environments is no longer an issue with deep neural networks.

5.7.11 When Not to Use Reinforcement Learning?

Although reinforcement learning model is a promising technique, it is not suitable for all situations. For example, you should not consider using reinforcement learning in the following situations:

1. When there is not enough data to solve the problem with a supervised learning method
2. When there is limited time and computing resources as reinforcement learning algorithms require heavy computing resources and are time-consuming.



PROGRAMMER'S ZONE

Perform k-nn on the iris data set

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
url = "https://archive.ics.uci.edu/ml/machine-learning-databases/
iris/iris.data"
# Assign column names to the dataset
names = ['sepal-length', 'sepal-width', 'petal-length', 'petal-
width', 'Class']
# Read dataset to pandas dataframe
dataset = pd.read_csv(url, names=names)
dataset.head()
# split our dataset into its attributes and labels
X = dataset.iloc[:, :-1].values # X contains first four columns of
dataset
y = dataset.iloc[:, 4].values# y has labels
from sklearn.model_selection import train_test_split
# to avoid over-fitting, create training and test splits
X_train, X_test, y_train, y_test = train_test_split(X, y, test_
size=0.20)
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
scaler.fit(X_train)
X_train = scaler.transform(X_train)
X_test = scaler.transform(X_test)
# use k-nn algorithm
from sklearn.neighbors import KNeighborsClassifier
classifier = KNeighborsClassifier(n_neighbors=7)
classifier.fit(X_train, y_train)
y_pred = classifier.predict(X_test)
# generating the confusion matrix
from sklearn.metrics import classification_report, confusion_matrix
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

OUTPUT

```
[[11 0 0]
[ 0 13 0]
[ 0 0 6]]
precision recall f1-score support
Iris-setosa 1.00 1.00 1.00 11
Iris-versicolor 1.00 1.00 1.00 13
Iris-virginica 1.00 1.00 1.00 6
accuracy 1.00 30
macro avg 1.00 1.00 1.00 30
weighted avg 1.00 1.00 1.00 30
```

Implementing Decision Tree Algorithm on Iris Data Set

```
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier
from sklearn.tree import export_text
iris = load_iris()
decision_tree = DecisionTreeClassifier(random_state=0, max_depth=2)
decision_tree = decision_tree.fit(iris.data, iris.target)
r = export_text(decision_tree, feature_names=iris['feature_names'])
print(r)
```

OUTPUT

```
|--- petal width (cm) <= 0.80
| |--- class: 0
|--- petal width (cm) > 0.80
| |--- petal width (cm) <= 1.75
| | |--- class: 1
| |--- petal width (cm) > 1.75
| | |--- class: 2
```

Implement Random Forest Algorithm on iris Data Set

```
#Import scikit-learn dataset library
from sklearn import datasets
#Load dataset
iris = datasets.load_iris()
# print the label species(setosa, versicolor,virginica)
print(iris.target_names)
# print the names of the four features
print(iris.feature_names)
# print the iris data (top 5 records)
print(iris.data[0:5])
# print the iris labels (0:setosa, 1:versicolor, 2:virginica)
print(iris.target)
# Creating a DataFrame of given iris dataset.
import pandas as pd
data=pd.DataFrame({
'sepal length':iris.data[:,0],
'sepal width':iris.data[:,1],
'petal length':iris.data[:,2],
'petal width':iris.data[:,3],
'species':iris.target
})
data.head()
# Import train_test_split function
from sklearn.model_selection import train_test_split
X=data[['sepal length', 'sepal width', 'petal length', 'petal
width']] # Features
y=data['species'] # Labels
# Split dataset into training set and test set
X_train, X_test, y_train, y_test = train_test_split(X, y, test_
size=0.3) # 70% training and 30% test
#Import Random Forest Model
from sklearn.ensemble import RandomForestClassifier
#Create a Gaussian Classifier
clf=RandomForestClassifier(n_estimators=100)
#Train the model using the training sets y_pred=clf.predict(X_test)
clf.fit(X_train,y_train)
y_pred=clf.predict(X_test)
#Import scikit-learn metrics module for accuracy calculation
from sklearn import metrics
# Model Accuracy, how often is the classifier correct?
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
```

OUTPUT

```
['setosa' 'versicolor' 'virginica']
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)',
'petal width (cm)']
[[5.1 3.5 1.4 0.2]
[4.9 3. 1.4 0.2]
[4.7 3.2 1.3 0.2]
[4.6 3.1 1.5 0.2]
[5. 3.6 1.4 0.2]]
[0 0 0 0 0 0 0……..1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 ……. 2 2 2 2 2 2]
Accuracy: 0.9555555555555556
```

Download the Wholesale customers data from https://archive.ics.uci.edu/ml/machine-learning-databases/00292/Wholesale%20customers%20data.csv and perform k-means++ algorithm for clustering. Use elbow method to decide the optimum number of k. Also use the inertial value to evaluate metrics.

```
# importing required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
# reading the data and looking at the first five rows of the data
data=pd.read_csv("Wholesale customers data.csv")
print(data.head())
# standardizing the data
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
data_scaled = scaler.fit_transform(data)
# statistics of scaled data
pd.DataFrame(data_scaled).describe()
# defining the kmeans function with initialization as k-means++
kmeans = KMeans(n_clusters=2, init='k-means++')
# fitting the k means algorithm on scaled data
kmeans.fit(data_scaled)
# inertia_ indicates how well the clusters are formed
print("K-means Interia = ",kmeans.inertia_)
```

OUTPUT

```
Channel Region Fresh Milk Grocery Frozen Detergents_Paper Delicassen
0   2       3    12669  9656    7561     214             2674            1338
1   2       3    7057   9810    9568     1762            3293            1776
2   2       3    6353   8808    7684     2405            3516            7844
3   1       3    13265  1196    4221     6404            507             1788
4   2       3    22615  5410    7198     3915            1777            5185
K-means Interia = 2599.384423783626
```

Use the different values for weather and temperature to predict whether a child should go out to play out or not using Naïve Bayes.

```
# Assigning features and label variables
weather=['Sunny','Sunny','Overcast','Rainy','Rainy','Rainy',
'Overcast','Sunny','Sunny',
'Rainy','Sunny','Overcast','Overcast','Rainy']
temp=['Hot','Hot','Hot','Mild','Cool','Cool','Cool','Mild','Cool','
Mild','Mild','Mild','Hot','Mild']
play=['No','No','Yes','Yes','Yes','No','Yes','No','Yes','Yes','Yes'
,'Yes','Yes','No']
# Import LabelEncoder
from sklearn import preprocessing
#creating labelEncoder
le = preprocessing.LabelEncoder()
# Converting string labels into numbers.
weather_encoded=le.fit_transform(weather)
print(weather_encoded)
# Converting string labels into numbers
temp_encoded=le.fit_transform(temp)
label=le.fit_transform(play)
print("Temp:",temp_encoded)
print("Play:",label)
#Combinig weather and temp into single listof tuples
features=tuple(zip(weather_encoded,temp_encoded))
print(features)
import numpy as np
features = np.asarray(features)
#Import Gaussian Naive Bayes model
from sklearn.naive_bayes import GaussianNB
#Create a Gaussian Classifier
model = GaussianNB()
# Train the model using the training sets
features.reshape(-1,1)
model.fit(features,label)
#Predict Output
predicted= model.predict([[0,2]]) # 0:Overcast, 2:Mild
print("Predicted Value:", predicted)
```

OUTPUT

```
[2 2 0 1 1 1 0 2 2 1 2 0 0 1]
Temp: [1 1 1 2 0 0 0 2 0 2 2 2 1 2]
Play: [0 0 1 1 1 0 1 0 1 1 1 1 1 0]
((2, 1), (2, 1), (0, 1), (1, 2), (1, 0), (1, 0), (0, 0), (2, 2),
(2, 0), (1, 2), (2, 2), (0, 2), (0, 1), (1, 2))
```

Write a program to implement Multi-layer Perceptron (MLP) on a simple data matrix

```
from sklearn.neural_network import MLPClassifier
X = [[0., 0.], [1., 1.]]
y = [0, 1]
clf = MLPClassifier(solver='lbfgs', alpha=1e-5,
hidden_layer_sizes=(5, 2), random_state=1)
clf.fit(X, y)
print(clf.predict([[2., 2.], [-1., -2.]]))
```

OUTPUT

[1 0]

Perform SVM on breast cancer data

```
#Import scikit-learn dataset library
from sklearn import datasets
#Load dataset
cancer = datasets.load_breast_cancer()
# print the names of the 13 features
print("Features: ", cancer.feature_names)
# print the label type of cancer('malignant' 'benign')
print("Labels: ", cancer.target_names)
# print data(feature)shape
print(cancer.data.shape)
# print the cancer data features (top 5 records)
#print(cancer.data[0:5])
# print the cancer labels (0:malignant, 1:benign)
# print(cancer.target)
# Import train_test_split function
from sklearn.model_selection import train_test_split
# Split dataset into training set and test set
X_train, X_test, y_train, y_test = train_test_split(cancer.data,
cancer.target, test_size=0.3,random_state=109) # 70% training and
30% test
#Import svm model
from sklearn import svm
#Create a svm Classifier
clf = svm.SVC(kernel='linear') # Linear Kernel
#Train the model using the training sets
clf.fit(X_train, y_train)
#Predict the response for test dataset
y_pred = clf.predict(X_test)
#Import scikit-learn metrics module for accuracy calculation
from sklearn import metrics
# Model Accuracy: how often is the classifier correct?
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
# Model Precision: what percentage of positive tuples are labeled
as such?
print("Precision:",metrics.precision_score(y_test, y_pred))
# Model Recall: what percentage of positive tuples are labelled as
such?
print("Recall:",metrics.recall_score(y_test, y_pred))
```

OUTPUT

```
Features: ['mean radius' 'mean texture' 'mean perimeter' 'mean
area'
'mean smoothness' 'mean compactness' 'mean concavity'
'mean concave points' 'mean symmetry' 'mean fractal dimension'
'radius error' 'texture error' 'perimeter error' 'area error'
'smoothness error' 'compactness error' 'concavity error'
'concave points error' 'symmetry error' 'fractal dimension error'
'worst radius' 'worst texture' 'worst perimeter' 'worst area'
'worst smoothness' 'worst compactness' 'worst concavity'
'worst concave points' 'worst symmetry' 'worst fractal dimension']
Labels: ['malignant' 'benign']
(569, 30)
Accuracy: 0.9649122807017544
Precision: 0.9811320754716981
Recall: 0.9629629629629629
```



KEY TERMS

Inference: It is the process of generating new information. It is an automatic process that has no goals.

Reasoning: The mental process of deriving logical conclusion and making predictions from available knowledge, facts and beliefs is called reasoning.

Deductive reasoning: **It is a** strategic approach that uses available facts, information or knowledge to draw valid conclusions.

Probability: It is a chance that an uncertain event will occur. It indicates the likelihood that an event will occur.

Event: Each possible outcome of a variable is called an event.

Sample space: The collection of all possible events is called sample space.

Conditional probability: It is the possibility of an event based on the existence of a previous event.

Joint Probability: It is the probability of event A and event B happening, P(A and B) together.

Bayesian network: It is a probabilistic graphical model which represents a set of variables and their conditional dependencies using a directed acyclic graph.

Machine learning: It is a branch of computer science that is specially dedicated to study and design algorithms that can produce predicates from the given dataset.

Clustering: It is the process of grouping together objects in a single cluster that are similar properties.

Cluster: It is a group of data objects that are similar to other objects in the same cluster and dis-similar to data objects in other clusters

Environment: It is a scenario in which the agent interacts and performs actions. The environment is assumed to be random in nature.

State (s): It refers to the current situation of the environment after an action is taken by the agent.

Policy (π): It is a strategy applied by the agent in the current state to decide the next action.

Value Function: **It is t**he function which returns the value of a state. This value returns the total amount of reward. The agent is expected to start from this state.

Model of the environment: It is a model that mimics the actual behaviour of the environment. It helps to draw inferences as it helps to observe how the environment will behave in a particular scenario.



RECAPITULATION POINTS

- The term ‘reasoning’ is associated with thinking, intellect and cognition that directs the process of generating one idea from the other.

- A reasoning system is targeted or is directed to achieve the goal of deducing an inference.
- Inductive reasoning, also known as cause-effect reasoning, generates hypothesis rather than drawing conclusion(s) to the facts at the beginning of the process.
- Common sense reasoning helps machines simulate human ability to make decisions and perceptions about real-world events.
- Monotonic reasoning uses facts, information and knowledge to draw a conclusion. Non-monotonic reasoning labels a conclusion as invalid if, on addition of new or more information, a previously generated conclusion becomes contradictory.
- Abductive reasoning starts with an incomplete set of facts, information and knowledge to find the most appropriate explanation and conclusion.
- Bayes' theorem determines the probability of an event with uncertain knowledge.
- Bayesian networks help us to effectively visualize the probabilistic model for each domain and study the relationship between random variables in the form of a graph.
- Bayesian networks satisfy the Local Markov property which states that a node is conditionally independent of its non-descendants (given its parents).
- A computer program is said to learn from experience 'E' with respect to some class of tasks 'T' and performance measure 'P', if its performance at tasks in 'T', as measured by 'P', improves with experience E.
- Rote learning is based on memorization, ignores the inner complexities and the learner just recalls the stored knowledge.
- Cognitive loading theory states that the amount of information and interactions that must be processed simultaneously can either underload or overload the finite working memory.
- Explanation based learning (EBL) uses an explicit domain theory to generate an explanation or proof of a training example.
- *Value-based algorithms* formulate optimal policy by estimating the value function of every state accurately. The aim is to maximize a value function V(s) as the agent is expecting a long-term return of the current states under policy π.



ANSWER THEM

1. Define reasoning.
2. Give the basic features of a reasoning system.
3. Write a short note on common-sense reasoning.
4. Differentiate between the following:
    1. Deductive and inductive reasoning
    2. Monotonic and non-monotonic reasoning
    3. Supervised and unsupervised learning
    4. Agglomerative and divisive clustering
    5. Reinforcement learning and supervised learning
    6. Positive and negative reinforcement
5. Explain the different types of reasoning.
6. Why do we need probabilistic reasoning?
7. What are Bayesian networks?
8. Give the main components of a learning system.
9. What does the cognitive loading theory state?
10. Discuss the categories of clustering algorithms.
11. What are the different components of reinforcement learning?
12. Give the merits and demerits of reinforcement learning.
13. Consider the following sentences in the knowledge base.
    Birds can fly. Penguins cannot fly. Pitty is a bird.
    Can you conclude that Pitty can fly?
    It becomes an example of non-monotonic reasoning. Justify this statement.
14. Identify the type of reasoning and explain it in detail.
    1. Premise: All of the pigeons we have seen in the zoo are white.
       Conclusion: Therefore, we can expect all the pigeons to be white.
    2. Implication: Cricket ground is wet if it is raining
       Axiom: Cricket ground is wet.
       Conclusion It is raining.
15. Write a short note on the following:
    1. Conditional probability
    2. Bayes' theorem
    3. Machine learning

4. Clustering
5. Explanation-based learning
6. Model free model reinforcement model

16. A box contains three coins: two regular coins and one fake two-headed coin (P(H)=1P(H)=1).
    1. You pick a coin at random and toss it. What is the probability that it lands heads up?
    2. You pick a coin at random and toss it, and get heads. What is the probability that it is the two-headed coin?
17. In a group of 10 people, 4 people bought apples, 3 bought oranges and 2 bought apples and oranges. If a buyer chose randomly bought apples, using the conditional probability formula, find the probability they also bought oranges.
18. My neighbour has 2 children. I learn that she has a son, Adam. What is the probability that Adam's sibling is a boy?
19. A fair die is rolled. What is the probability of A given B where A is the event of getting an even number and B is the event of getting a number less than or equal to 2?
20. A die is rolled twice and two numbers are obtained. Let X be the outcome of the first role and Y be the outcome of the second roll. Given that X + Y = 5, what is the probability of X = 4 or Y = 4?
21. 70% of your friends like chocolate and 35% like chocolate and strawberry. What percent of those who like chocolate also like strawberry?
22. Two dies are thrown simultaneously, and the sum of the numbers obtained is found to be 7. What is the probability that the number 3 has appeared at least once?
23. In a group of 100 computer buyers, 40 bought CPUs, 30 purchased monitors and 20 purchased CPUs and monitors. If a computer buyer chose at random and bought a CPU, what is the probability they also bought a monitor?
24. In a group of 100 sports car buyers, 40 bought alarm systems, 30 purchased bucket seats and 20 purchased an alarm system and bucket seats. If a car buyer chosen at random bought an alarm system, what is the probability they also bought bucket seats?

25. Consider the contingency table given below.

	Have Pets	Do Not Have Pets	Total
Male	0.41	0.08	0.49
Female	0.45	0.06	0.51
Total	0.86	0.14	1

What is the probability that a randomly selected person is male, given that they own a pet?

26. Harry installed a new burglar alarm at his home to detect burglary. The alarm reliably responds at detecting a burglary but also responds for minor earthquakes. Harry has two neighbours—David and Sophia—who have taken responsibility to inform Harry at work when they hear the alarm. David always calls Harry when he hears the alarm, but sometimes, he got confused with the phone ringing and calls at that time too. On the other hand, Sophia likes to listen to high music, so sometimes, she misses to hear the alarm. Here, we would like to compute the probability of the burglary alarm.



FILL UP

Calculate the probability that the alarm has sounded, but there is neither a burglary, nor an earthquake occurred, and David and Sophia both called Harry.

1. _____ is a strategic approach that uses available facts, information or knowledge to draw valid conclusions.
2. Inductive reasoning generates _____.
3. Inductive and deductive reasoning are forms of _____ logic.
4. Common-sense reasoning is gained through _____.
5. Common sense works with _____ rules.
6. Vivekananda said, ‘In a conflict between the heart and the brain, follow your heart’. is an example of _____ reasoning.
7. _____ reasoning is used in conventional reasoning logic-based system.
8. _____ reasoning is very useful for robot navigation.

9. In AI, we use _____ to indicate the uncertainty in knowledge.
10. _____ indicates the likelihood that an event will occur.
11. Each possible outcome of a variable is called an _____.
12. The collection of all possible events is called _____.
13. If A and B are independent events, then conditional probability is given by _____.
14. _____ theorem determines the probability of an event with uncertain knowledge.
15. _____ networks help us to effectively visualize the probabilistic model for each domain.
16. In a Bayesian network, each _____ represents a random variable which can either be _____ or _____.
17. Bayesian Networks satisfy the _____ property which states that a node is conditionally independent of its parents.
18. _____ element of a learning system tells the learning element how the agent is doing.
19. _____ is the most widely used reinforcement learning algorithm.
20. _____ learning is usually done when there is no labelled data.
21. Clustering which is an example of _____ learning.
22. _____ is a group of data objects that are similar to other objects in the same group and dissimilar to objects in another group.
23. Two examples of partitional clustering algorithms are _____ and _____.
24. The _____ is used to prune all irrelevant aspects of the training example.
25. _____ function returns the total amount of reward in reinforcement learning.
26. In reinforcement learning, the agent continues to learn until it finds the best solution that gives the maximum _____.
27. In _____ model, for any state, the same action is produced by the policy.
28. RL algorithms work well in _____ and _____ environments.



STATE TRUE OR FALSE

1. A reasoning system is not modified externally by a programmer.
2. A reasoning system cannot solve new and unusual problems.
3. A reasoning system improves using hit-and-trial technique.
4. Prediction is done by pattern matching.
5. Inductive reasoning starts from the general premises to get to specific conclusion(s).
6. In deductive reasoning, the arguments can be valid or invalid based on the rules.
7. Addition of new facts and knowledge in the database cannot alter the conclusion drawn at an earlier point in time.
8. Human’s perceptions for various things in daily life is an example of non-monotonic reasoning.
9. Monotonic reasoning labels a conclusion as invalid on addition of new or more information.
10. In first-order logic and propositional logic, knowledge is represented with certainty.
11. The unconditional probability of an event P(A) is known as joint probability.
12. Posterior probability P(A|B) is read as probability of hypothesis A when an evidence B has occurred.
13. The Bayesian network graph may contain any cycles.
14. Machine learning is dedicated to study and design algorithms that can produce predicates from the given dataset.
15. Critics make improvements in the system by accepting knowledge about performance element and feedbacks.
16. While instance-based learning is data-driven, EBL is knowledge-driven.
17. In reinforcement learning, the system knows the desirable outcomes but does not know which actions should be performed to get those outcomes.
18. Chunking is less efficient than normal memorizing.
19. People learn better when a complex continuous lesson is broken into separate segments.

20. Unsupervised machine learning algorithm uses an inference function to train it using labelled training data.
21. Objects in a cluster are similar to each other and objects in different clusters vary significantly from each other.
22. Clustering is better than classification.
23. Divisive clustering follows a bottom-up approach.
24. Density-based approach does not require the user to specify the number of clusters.
25. EBL is also known as inductive learning.
26. In reinforcement learning, the environment is assumed to be random in nature.
27. Reinforcement learning should be used when the environment is known, but an analytic solution is not known.
28. Reinforcement learning divides the problem into subproblems.



MULTIPLE CHOICE QUESTIONS

1. _____ system is targeted or is directed to achieve the goal of deducing an inference.
    1. Knowledge representation
    2. Intelligent
    3. Reasoning
    4. Prediction
2. _____ is the mental process of deriving a logical conclusion and making predictions from available knowledge, facts and beliefs.
    1. Knowledge representation
    2. Intelligent
    3. Reasoning
    4. Prediction
3. In _____ reasoning, every result is based on the facts and ideas and is drawn using a top-down approach.
    1. Deductive
    2. Inductive
    3. Common sense
    4. None of these

4. Reasoning allows AI technologies to extract critical information from large sets of _____ data.
    1. Structured
    2. Unstructured
    3. Both of these
    4. None of these
5. Identify the incorrect statement.
    1. Deductive reasoning usually starts from the general premises to get to specific conclusion(s).
    2. Inductive reasoning does not draw conclusion(s).
    3. Inductive reasoning is preferred in cases with a limited set of facts.
    4. Deductive reasoning is fast and easy.
6. _____ reasoning helps machines to simulate human ability to make decisions and perceptions about real-world events.
    1. Deductive
    2. Inductive
    3. Common sense
    4. None of these
7. Any theorem proving is an example of _____ reasoning.
    1. Deductive
    2. Inductive
    3. Common sense
    4. Monotonic
8. _____ reasoning is mainly used in conventional reasoning systems and logic-based systems.
    1. Deductive
    2. Inductive
    3. Common sense
    4. Monotonic
9. Identify the incorrect statement about monotonic reasoning.
    1. Each proof will remain valid for a short duration.
    2. It cannot express real-world scenarios.
    3. Hypothesis knowledge cannot be expressed.
    4. New knowledge from the real world cannot be added.
10. _____ reasoning cannot be used for theorem proving.
    1. Deductive
    2. Non-monotonic

3. Common sense
4. Monotonic

11. Identify the incorrect statement about abductive reasoning.
    1. Conclusions are based on facts known at the present.
    2. It does not give any importance to outdated facts and information.
    3. It does not rectify the plausible conclusion.
    4. None of these
12. Probabilistic reasoning can be applied when _____.
    1. There are unpredictable outcomes.
    2. Specifications or possibilities of predicates become too large to handle.
    3. An unknown error occurs during an experiment.
    4. All of these
13. Bayesian network is based on _____ probability.
    1. Joint
    2. Conditional
    3. Both of these
    4. None of these
14. _____ element is the agent that acts in the world.
    1. Performance
    2. learning
    3. critic
    4. problem generator
15. Human learners learn a poem by _____ learning.
    1. Rote
    2. Induction
    3. Explanation Based
    4. Reinforcement
16. _____ learning is also known as learning by example.
    1. Rote
    2. Induction
    3. Explanation-based
    4. Reinforcement
17. Identify the incorrect statement about learning by taking advice.
    1. Derives inference rather than just doing rote learning.
    2. With more stored knowledge, the reliability of the knowledge source improves.

3. It is based on memorization.
4. Programs are written to perform specific tasks.

18. _____ learning uses rewards and punishments to train the agent.
    1. Rote
    2. Induction
    3. Explanation Based
    4. Reinforcement
19. Identify the incorrect statement about partial clustering.
    1. Works well when clusters have a spherical shape.
    2. Scalable with respect to algorithm complexity.
    3. Performs well with complex cluster shapes having different sizes.
    4. Does not work with clusters of different densities.
20. OPTICS is a popular example of _____ clustering algorithms.
    1. Partitional
    2. Hierarchical
    3. Density-based
    4. Agglomerative
21. Identify the incorrect statement about density-based clustering technique.
    1. Works well when clusters are of non-spherical shapes.
    2. Performs well even on a data set having noise or outliers.
    3. Difficult to identify clusters of varying densities.
    4. Well suited for clustering in high-dimensional spaces.
22. _____ contains a set of rules that describe relationships between objects and actions in a domain.
    1. Domain theory
    2. training example
    3. goal concept
    4. operational criterion
23. The _____ perceives and explore the environment before acting upon it.
    1. Input
    2. Output
    3. Agent
    4. Environment
24. Identify the incorrect statement about reinforcement learning.

1. There is no supervisor. Only rewards direct the actions to achieve outcomes.
2. Sequential decision-making
3. It is based on hit and trial method.
4. It cannot achieve artificial general intelligence (AGI).

25. If I put my hand in fire, it will burn. This is an example of _____ reasoning.
    1. Deductive
    2. Non-monotonic
    3. Common sense
    4. Monotonic
26. The Earth revolves around the Sun. This is an example of _____ reasoning.
    1. Deductive
    2. Non-monotonic
    3. Common sense
    4. Monotonic



ANSWERS

Fill Up

1. Deductive reasoning
2. Hypothesis
3. propositional
4. experiences
5. heuristic
6. Monotonic.
7. Monotonic
8. Non-monotonic
9. Probability
10. Probability
11. event
12. sample space
13. P(A | B) = P (A)

14. Bayes
15. Bayesian
16. Node, continuous, discrete
17. Local Markov
18. critic
19. Q-Learning
20. Unsupervised
21. Unsupervised
22. cluster
23. *k*-means and *k*-medoids.
24. domain theory
25. Value
26. reward.
27. *Deterministic*
28. dynamic and uncertain

State True or False

1. True
2. False
3. True
4. True
5. False
6. True
7. True
8. True
9. False
10. True
11. False
12. True
13. False
14. True
15. False
16. True
17. True
18. False
19. True

20. False
21. True
22. True
23. False
24. True
25. False
26. True
27. True
28. False

Multiple Choice Questions

1. c
2. c
3. a
4. c
5. d
6. c
7. d
8. d
9. a
10. b
11. d
12. d
13. c
14. a
15. a
16. b
17. c
18. d
19. c
20. c
21. d
22. a
23. c
24. d
25. c

26. d